

SSC Maharashtra Board — Science and Technology (Part II)

March 2026 — Model Solutions

Time: 2 Hours

Max. Marks: 40

Q.1

Q.1 (A) Choose the correct alternative: [5 Marks]

(i) The causality behind the sudden changes was understood due to ... principle of Hugo de Vries.

(A) Appendix (B) Mutation (C) Transcription (D) Translocation

Ans: (B) Mutation

(ii) After complete oxidation of a glucose molecule, ... number of ATP molecules are formed.

(A) 18 (B) 28 (C) 38 (D) 48

Ans: (C) 38

(iii) Bat is included in class ...

(A) Amphibia (B) Reptilia (C) Aves (D) Mammalia

Ans: (D) Mammalia

(iv) In Maharashtra the major hydroelectric power plant is located at ...

(A) Chandrapur (B) Koyana (C) Parali (D) Tarapur

Ans: (B) Koyana

(v) A first state to start a separate cyber crime unit in our country is ...

(A) Madhya Pradesh (B) Maharashtra (C) Gujrat (D) Karnataka

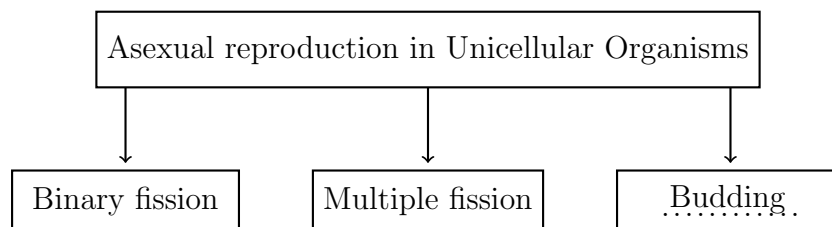
Ans: (B) Maharashtra

Q.1 (B) Answer the following questions:

[5 Marks]

(i) Fill in the box with proper words:

Asexual reproduction in Unicellular Organisms



(ii) Which day is observed as World Biodiversity Day?

World Biodiversity Day is observed on **22nd May** every year.

(iii) Identify me, who am I?

"I am the first clone sheep born in Scotland."

Dolly

(iv) Write the molecular formula of acetic acid.

Acetic acid has the molecular formula:

CH₃COOH

(v) State whether the following statement is true or false:

"The chairman of National Disaster Management is Sarpanch."

False

Q.2

Q.2 (A) Give scientific reasons (any *two*):

[4 Marks]

(i) The popularity of probiotic products is increasing.

1. Probiotic products contain **live beneficial microorganisms** (especially bacteria like *Lactobacillus* and *Bifidobacterium*) which are good for human health.
2. These beneficial bacteria help in **improving digestion** by maintaining a healthy balance of gut flora (microbiome) in the intestine.
3. They help in boosting the **immune system** and preventing harmful bacteria from growing in the intestine.

4. Probiotic products like curd, yogurt, and fermented foods help in absorption of nutrients, synthesis of vitamins, and prevention of diseases like diarrhoea and allergies.
5. Growing awareness about **preventive healthcare** and the benefits of good gut bacteria has increased the demand for probiotic products.

∴ The popularity of probiotic products is increasing.

(ii) Body temperature of reptiles is not constant.

1. Reptiles are **cold-blooded animals** (poikilotherms/ectotherms).
2. Their body temperature **changes with the surrounding environmental temperature**. They do not have an internal mechanism to regulate their body temperature.
3. They lack efficient **thermoregulatory mechanisms** like sweating or shivering that are present in warm-blooded animals (endotherms).
4. Reptiles depend on external sources of heat, such as sunlight, to warm their body. This is called **basking behaviour**.
5. Since the environmental temperature varies throughout the day and across seasons, the body temperature of reptiles also fluctuates accordingly.

∴ Body temperature of reptiles is not constant.

(iii) Nowadays farmers are opting for organic farming.

1. **Chemical fertilizers and pesticides** used in conventional farming cause soil degradation, water pollution, and harm to beneficial organisms.
2. Organic farming uses **natural fertilizers** (compost, vermicompost, green manure) and **bio-pesticides**, which are eco-friendly and do not contaminate soil and water.
3. Food produced through organic farming is **free from harmful chemical residues**, making it safer and healthier for consumption.
4. Organic farming helps maintain **soil fertility** and biodiversity in the long run.
5. There is growing **consumer demand** for organic food products due to increasing health awareness, which gives better market prices to farmers.
6. Organic farming is **sustainable** and helps in conservation of natural resources.

∴ Nowadays farmers are opting for organic farming.

Q.2 (B) Answer the following questions (any *three*): [6 Marks]

(i) What is meant by carbon dating? Give any *one* use of this technique.

Carbon dating (Radiocarbon dating):

Carbon dating is a scientific technique used to determine the **age of fossils, archaeological artefacts, and other objects** that contain organic material.

- All living organisms contain the radioactive isotope **Carbon-14 (C-14)** along with the stable isotope Carbon-12 (C-12).
- When an organism dies, it stops absorbing carbon. The C-14 in the dead organism begins to **decay at a known rate** (half-life of approximately 5,730 years).
- By measuring the **ratio of C-14 to C-12** remaining in a fossil or artefact, scientists can calculate how long ago the organism died.

Use of carbon dating:

Carbon dating is used to determine the **age of fossils** found in excavations, helping scientists understand the evolutionary history and timeline of organisms that lived in the past.

(ii) Complete the table:

Sr. No.	Protein	Name of Organ
1		Bones
2	Insulin	

Ans:

Sr. No.	Protein	Name of Organ
1	Ossein	Bones
2	Insulin	Pancreas

(iii) Read the paragraph and answer the questions:

“Paddy is cultivated on large scale in various states of South India. Paddy fields are frequently attacked by grasshoppers. Similarly, frogs are also present in large number in the mud of paddy fields, to feed upon grasshoppers and snakes are also present therein to feed upon their favourite food-frogs. However, if frog population declines all of a sudden,”

(a) What will be the effect on paddy crop?

If the frog population declines suddenly:

- Frogs are the natural predators of grasshoppers. Without frogs, the **grasshopper population will increase** uncontrolled.
- The increased number of grasshoppers will cause **severe damage to the paddy crop** by feeding on it excessively.

- This will lead to **crop loss and reduced yield**.

The paddy crop will be severely damaged due to increase in grasshoppers.

(b) Number of which components will increase?

If frog population declines then then number of **grasshoppers** will increase (as their predator, frog, is absent).

(iv) Write any *two* advantages of hydroelectric power generation.

1. Hydroelectric energy is renewable.
2. It is eco-friendly.

(v) The following are the pictures of some disasters. How will be your post disaster management in case you face any of those disasters?



(A)



(B)

(A) Post-disaster management for Gas leak:

1. Immediately **evacuate the affected area** and move to a safe location upwind.
2. **Cover nose and mouth** with a wet cloth to avoid inhaling toxic gases.
3. **Call emergency services** (fire brigade, ambulance, and disaster management authorities) immediately.

(B) Post-disaster management for Flood:

1. **Search and rescue** trapped or injured persons with the help of trained personnel.
 2. Provide **first aid and medical help** to the injured.
 3. Arrange for **temporary shelters, food, and clean drinking water** for affected families.
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Q.3 Answer the following questions (any *five*): [15 Marks]

(i) Explain Lamarckism with two examples.

Lamarckism: Jean-Baptiste Lamarck proposed that morphological changes occurring in living organisms are responsible for evolution and the reason behind those morphological changes is activities or laziness of that organism. He called this concept as principle of 'use or disuse of organs'.

Examples:

1. Long neck of the giraffe:

- According to Lamarck, the ancestors of giraffes had short necks and short forelimbs.
- Due to scarcity of food on the ground, they had to **stretch their necks to reach leaves on tall trees**.
- Over generations of continuous stretching, their necks and forelimbs became progressively longer.
- This acquired character was inherited by successive generations, resulting in the modern long-necked giraffe.

2. Strong arms of a blacksmith:

- A blacksmith continuously uses his arms and shoulders while hammering iron.
- Due to constant use, his arm and shoulder muscles become very strong and well developed.
- According to Lamarck, such **acquired characteristics produced by use of organs** are inherited by offspring.
- Therefore, the children of the blacksmith would inherit stronger arms.

(ii) Write the type of cells in which mitosis occurs. Write four stages of karyokinesis.

Ans:

Mitosis occurs in **somatic cells (body cells)** and **stem cells**.

Four stages of Karyokinesis (nuclear division) in Mitosis:

1. **Prophase:**
2. **Metaphase:**
3. **Anaphase:**
4. **Telophase:**

(iii) Give two examples of each of the threatened species given below:

(a) Endangered species (b) Rare species (c) Vulnerable species

Ans:

(a) Endangered species:

Examples: Lion-tailed Monkey, Lesser Florican

(b) Rare species:

Example: Red Panda, Musk Deer

(c) Vulnerable species:

Example: Tiger, Lion

(iv) Which fuel is used in thermal power plant ? What are the problems associated with this type of power generation?

Ans: The main fuel used in thermal power plants is **coal**.

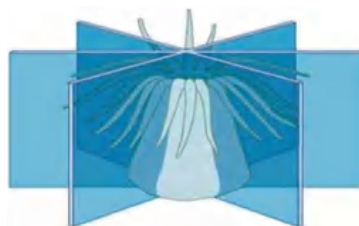
Problems associated with thermal power generation:

1. Air Pollution: Burning coal releases harmful gases like carbon dioxide, sulfur oxides, and nitrogen oxides, which can harm living beings.
2. Health Issues: The release of soot particles into the air can cause serious respiratory problems.

(v) Identify the type of body symmetry of the animals shown in the given pictures:



(A)



(B)



(C)

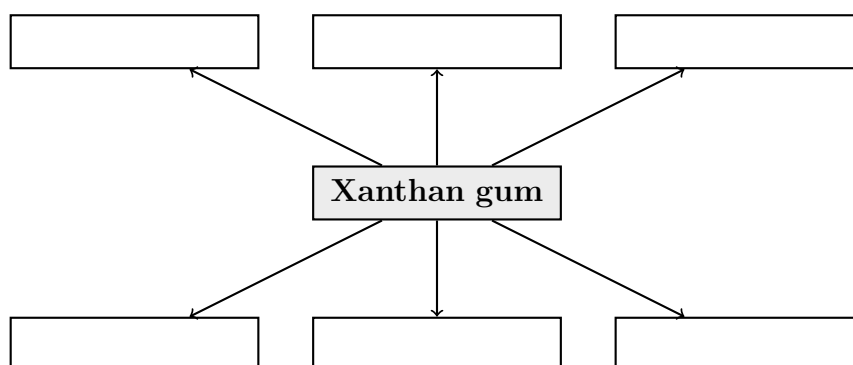
Ans:

(A) Asymmetrical body

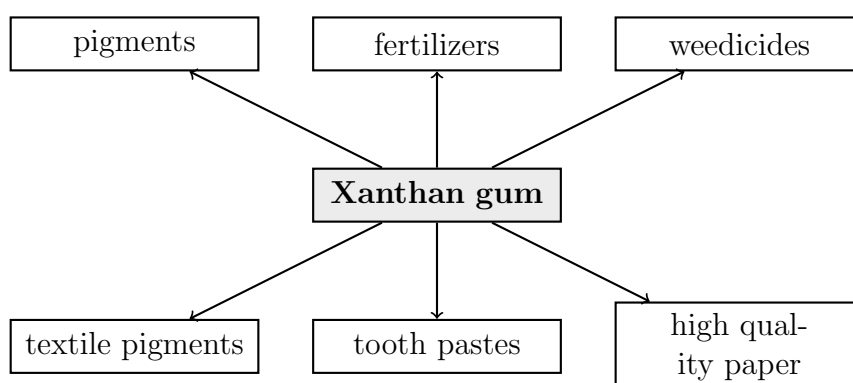
(B) Radial symmetry

(C) Bilateral symmetry

(vi) Complete the following conceptual picture with respect to uses:



Ans:



(vii) Which precautions will you take during spraying of pesticides?

The following precautions should be taken during spraying of pesticides:

1. **Wear protective clothing:** Always wear full-sleeved clothes, gloves, boots, goggles, and a face mask/respirator to avoid direct contact with pesticides.
2. **Cover nose and mouth:** Use a proper mask or a wet cloth to prevent inhalation of pesticide fumes and droplets.
3. **Spray in the direction of wind:** Always spray in the direction of the wind so that the pesticide spray does not blow back towards you.
4. **Wash hands and body:** After spraying, thoroughly wash hands, face, and other exposed body parts with soap and clean water. Change clothes immediately.
5. **Do not eat or drink:** Avoid eating or drinking during spraying to prevent accidental ingestion of pesticides.
6. **Keep children and animals away:** Ensure that children, pets, and livestock are kept away from the spraying area.
7. **Proper storage:** Store pesticides in original containers with labels, in a locked place away from food items and the reach of children.
8. **Follow instructions:** Read and follow the instructions on the pesticide label regarding dosage, dilution, and application method.

(viii) Write the names of any six materials in a first-aid box.

A first-aid box should contain the following essential materials:

1. Bandage strips of different size
 2. Medicated cotton
 3. Clean and dry cloth pieces
 4. Soap
 5. Antiseptic (Dettol/ Savlon)
 6. Blade
 7. Scissor
 8. Needle
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Q.4 Answer the following questions (any *one*): [5 Marks]

(i) (a) Write the names of any six factors affecting social health:

(b) What will you do ? Why ?

Your brother studying in XII has developed the stress.

Ans:

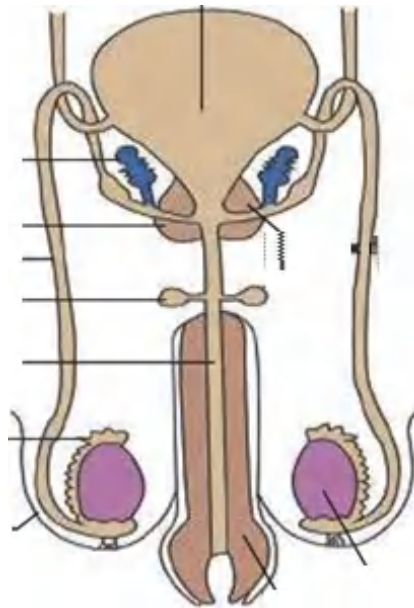
(a) The following factors affect social health:

1. Education
2. Financial Status
3. Job Opportunities
4. Social safety
5. Social treatment
6. Political conditions (views)
7. Social environment of the surrounding
8. Residential area

(b) If my brother is studying for XII board exams has developed stress, I will take the following steps to help him:

1. **Talk and support:** I will talk to my brother calmly and listen to his problems related to studies or exams and try to support him emotionally.
2. **Help in planning studies:** I will help him prepare a **proper study timetable** with study time, short breaks, and relaxation.
3. **Encourage exercise and relaxation:** I will encourage him to do **physical activities** like walking, yoga, or meditation to reduce stress.
4. **Ensure proper diet and sleep:** I will make sure he gets a **balanced diet and sufficient sleep** so that he can study with a fresh and healthy mind.

(ii) Observe the diagram given below and answer the questions:



(1) Write the name of the reproductive system in the above diagram. [1 Mark]

Male Reproductive System

(2) Name any *two* glands in the above diagram. [$\frac{1}{2} + \frac{1}{2} = 1$ Mark]

1. Prostate gland 2. Cowper's gland

(3) Mention any *one* duct in the above system. [1 Mark]

Ejaculatory duct

(4) What is the length of a sperm? [1 Mark]

Length of a sperm $\approx 60 \mu\text{m}$ (micrometres)

(5) Which hormone is secreted by the testes? [1 Mark]

Testosterone
